

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-	Team No-	Project Title- Airline Disruption Management UI
Team Lead Roll Number	2520030082	
Team Lead Name	K. Bhavana	
Team Member Roll Number	2520030490	
Team Member Name	C. Sri Dhruthi	

1. Abstract

The Airline Disruption Management UI is a frontend-based application developed to assist airline operations staff in handling flight disruptions such as delays and cancellations in an organized and efficient manner. In real-world scenarios, such disruptions often lead to confusion, delays in decision-making, and poor passenger experience. This project aims to provide a centralized dashboard where staff can quickly view disrupted flights, identify affected passengers, and take necessary actions without relying on multiple systems.

The proposed system offers features such as passenger re-accommodation, generation of hotel and meal vouchers, priority-based passenger handling, and mock communication updates. Although the application does not perform real-time bookings or live messaging, it simulates airline operations using interactive user interfaces and local data storage. This project demonstrates how a well-designed UI can simplify complex operational workflows and improve overall service management during disruptions.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Disruption Dashboard	2520030490	
Passenger Re-accommodation	2520030490	
Hotel Voucher Module	2520030490	
Meal Voucher Module	2520030490	
Communication Module	2520030490	
Status Board	2520030082	
Priority Queue	2520030082	
Export Module	2520030082	
History Logs	2520030082	
Admin Panel	2520030082	

3. Module Details

The project is divided between two team members to ensure balanced work distribution and efficient development of the Airline Disruption Management UI. Each member is responsible for designing, implementing, and testing their assigned modules, while both collaborate to integrate the complete system.

Member 1 handles the core operational modules of the system. This includes the Disruption Dashboard, where all delayed or cancelled flights are displayed in a structured format for quick monitoring. The Passenger Re-accommodation module allows reassignment of affected passengers to alternative flights based on availability. The Status Board module is responsible for displaying updated flight information in a simple and readable format. Additionally, the Priority Queue module manages passengers based on urgency levels such as high, medium, and low priority, ensuring that critical cases are handled first.

Member 2 focuses on passenger service and management-related modules. The Hotel Voucher module is responsible for generating accommodation details for passengers affected by long delays. The Meal Voucher module provides functionality to create meal vouchers with assigned values. The Communication module simulates sending notifications to passengers regarding flight updates. The Export module allows downloading system data in formats like CSV for reporting purposes. The History Logs module maintains a record of all actions performed within the system for tracking and analysis. Finally, the Admin Panel module enables management of flight and passenger data within the application.

Both members work together to ensure proper integration of all modules, maintain consistent user interface design, and provide a smooth and user-friendly experience throughout the application.

4. Frameworks / Platforms / Software Used

Frontend: HTML5, CSS3, JavaScript (ES6+), React.js, Bootstrap / Tailwind CSS

Backend: Node.js, Express.js, npm

API Testing: Postman

Database: MySQL (optional), LocalStorage

Libraries: Axios, React Router DOM, Chart.js / Recharts, React Hook Form, Day.js / Moment.js

Development Tools: VS Code, Git, Chrome DevTools, ESLint, Prettier

Deployment: Vercel / Netlify (optional), Localhost

Guide Name and Signature (remarks if any) _____